

E-Commerce Sales Analysis Dashboard

ReadyNest Data Analytics Internship — Week 1 Project

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Programme	B.Tech (2nd Semester)
Institute	Chandigarh Engineering College, Jhanjeri, Mohali
Project	ReadyNest Week 1 — Data Analytics
Dataset	E-Commerce Orders (500 records)
Tools	Python, MySQL, Microsoft Excel

■1829.44K Total Sales	■3658.89 Avg Order Value	500 Total Orders	■415.40K Total Profit
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1. Project Overview

This project is part of the **ReadyNest Data Analytics Internship — Week 1**. The objective was to perform end-to-end data analysis on an E-Commerce Orders dataset using **Python, MySQL, and Microsoft Excel**. The dataset contains 500 cleaned transaction records covering five product categories across four regions of India. The complete analytics pipeline — from raw data ingestion to visual dashboard — has been implemented.

1.1 Objectives

- Load and manage dataset using MySQL.
- Clean and preprocess raw transaction data (remove duplicates, handle nulls).
- Perform descriptive statistical analysis across categories and regions.
- Conduct univariate (histogram) and bivariate (cross-segment) analysis.
- Build a multi-panel visual dashboard using Python Matplotlib.
- Generate actionable business insights from sales trends.
- Present findings in a professional PDF report.

1.2 Technologies Used

Technology	Purpose
Python (Pandas, Matplotlib)	Data Cleaning, EDA, Visualization
MySQL / SQL	Database Management & Querying
Microsoft Excel	Raw & Cleaned Data Storage
Git & GitHub	Version Control & Project Hosting

2. Dataset Information

The E-Commerce Orders dataset contains sales transaction records for an online retail platform. The raw file had 520 rows including 12 duplicate records and 8 rows with missing Sales values.

Column	Description	Type
Order_ID	Unique order identifier	String
Order_Date	Date of transaction (YYYY-MM-DD)	Date
Category	Product category (5 types)	String
Ship_Mode	Shipping method used	String
Region	Geographic region (4 zones)	String
Quantity	Number of units ordered	Integer
Sales	Total sale amount (INR)	Float
Discount	Discount applied (0.0 - 0.3)	Float
Profit	Profit earned on order	Float

2.1 Data Cleaning Process

- Removed 12 duplicate records identified by exact row match.
- Dropped 8 rows with missing/null Sales values.
- Standardised column names (snake_case, no spaces).
- Verified and enforced correct data types for all columns.
- Converted Sales and Profit to float with 2 decimal precision.
- Formatted Order_Date uniformly to YYYY-MM-DD.
- Sorted dataset chronologically by Order_Date.
- Final cleaned dataset: **500 records, 0 nulls, 0 duplicates.**

3. Descriptive Statistics

Metric	Value
Total Sales	INR 1,829,444.63
Average Sales	INR 3,658.89
Total Orders	500
Total Profit	INR 415,404.69
Profit Margin	22.7%

3.1 Sales by Product Category

Category	Total Sales (INR)	Orders	Avg Sale (INR)	% Share
Electronics	686,341.00	176	3,899.66	37.5%
Furniture	862,269.64	98	8,798.67	47.1%
Clothing	164,026.19	130	1,261.74	9.0%
Sports	91,329.02	39	2,341.77	5.0%
Books	25,478.78	57	447.00	1.4%

4. Dashboard Visualisations

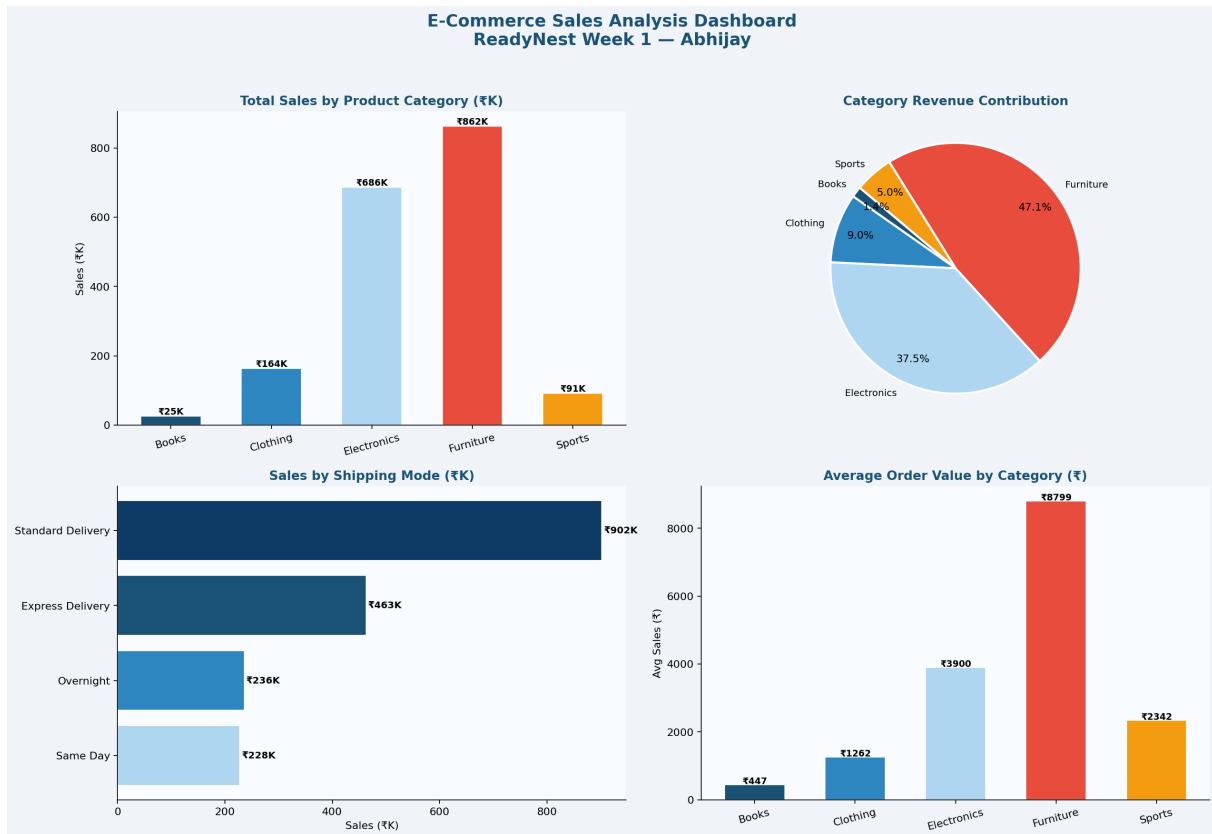


Figure 1: Main Sales Analysis Dashboard (4-panel)

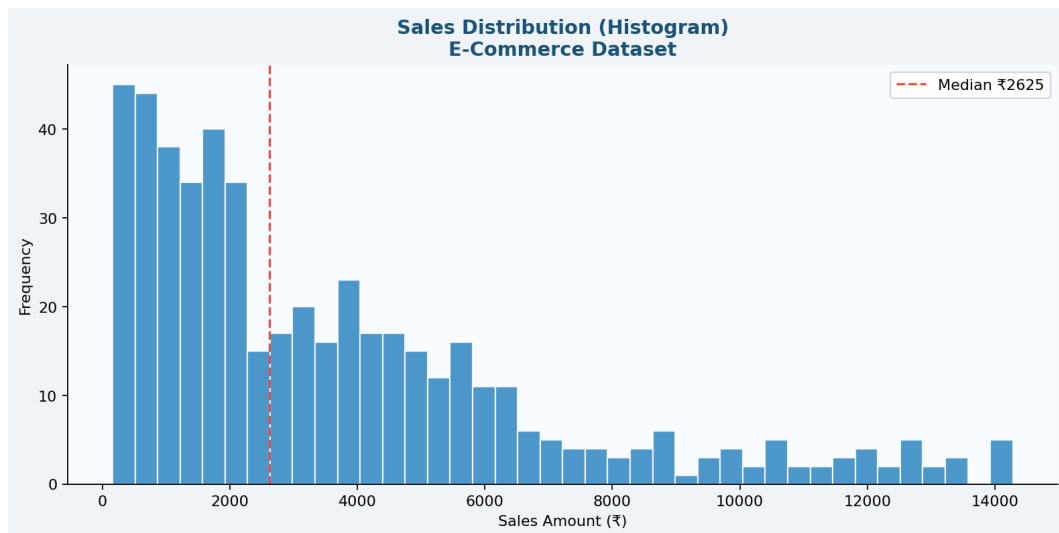


Figure 2: Sales Distribution Histogram

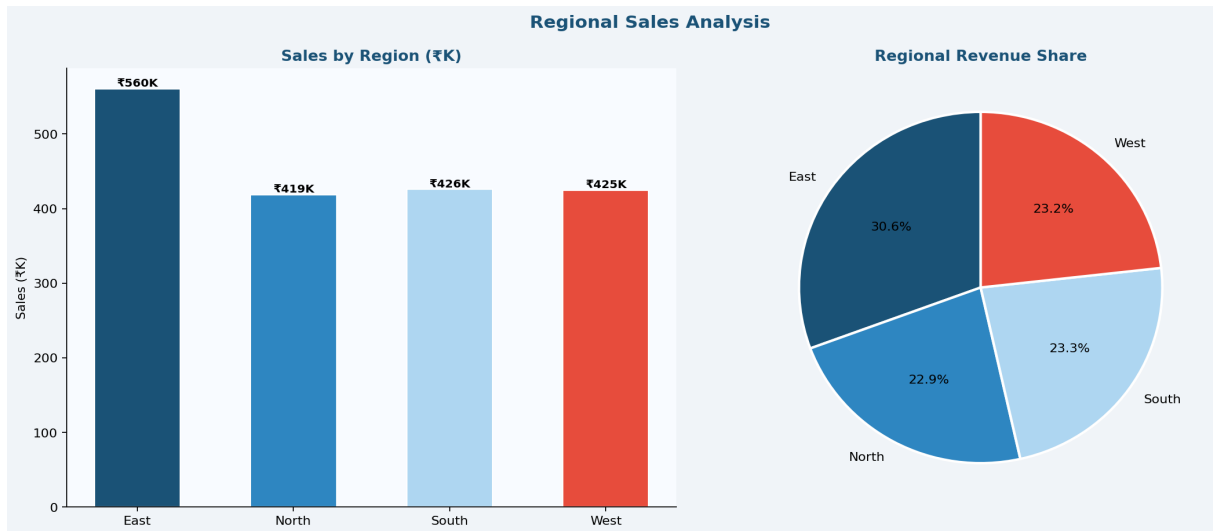


Figure 3: Regional Sales Analysis

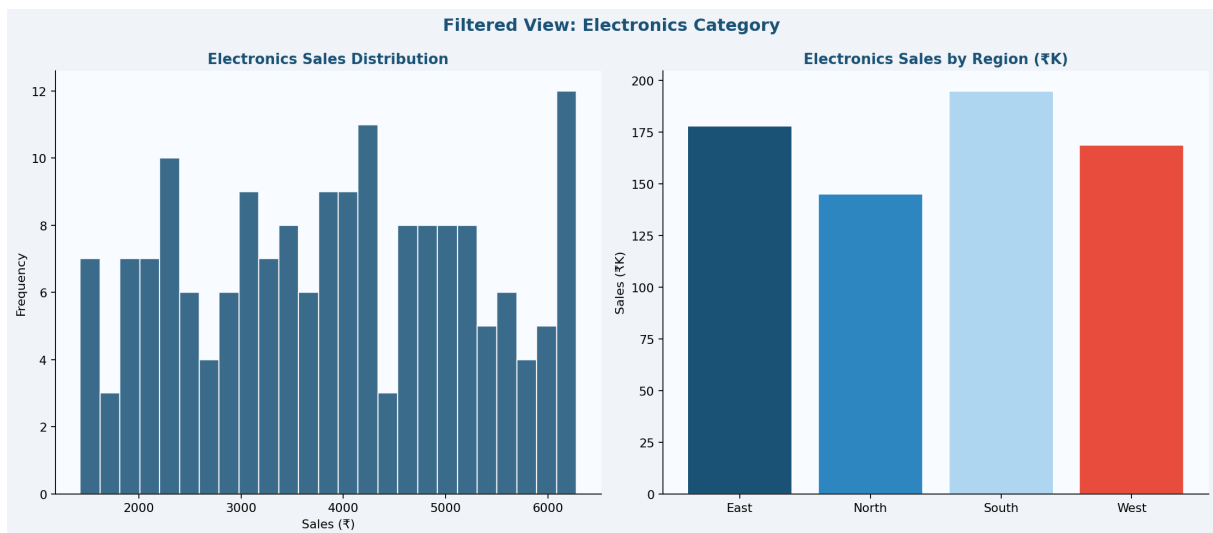


Figure 4: Filtered Dashboard — Electronics Category

5. Exploratory Data Analysis

5.1 Univariate Analysis — Sales Distribution

The histogram (Figure 2) shows that the sales distribution is **heavily right-skewed**. The majority of transactions fall in the low-to-mid sales range, while a small number of high-value Furniture orders create a long tail. This skew is typical in e-commerce datasets where most orders are routine purchases but occasional bulk or premium orders inflate the mean.

5.2 Bivariate Analysis

Sales by Product Category:

Category	Total Sales	Orders	Notes
Electronics	INR 6.86L	176	High volume, mid avg value
Furniture	INR 8.62L	98	Low volume, highest avg (INR 8,799)
Clothing	INR 1.64L	130	Moderate volume and value
Sports	INR 0.91L	39	Least orders overall
Books	INR 0.25L	57	Lowest avg sale (INR 447)

6. Key Business Insights

Insight 1: Furniture drives maximum revenue

Although Furniture has only 98 orders (19.6% of total), it generates the highest total sales of INR 8.62 Lakh (47.1% of revenue). Its average order value of INR 8,799 is the highest across all categories.

Insight 2: Electronics dominates order volume

Electronics accounts for 176 orders (35.2%) with INR 6.86 Lakh in sales. Strong volume + moderate average makes it the primary growth driver.

Insight 3: Standard Delivery is the preferred ship mode

Standard Delivery accounts for the largest shipping revenue share (~49%), followed by Express. Same Day and Overnight combined contribute under 25%.

Insight 4: East region leads in total revenue

The East region generates INR 5.60 Lakh, the highest among all four regions. All four regions are relatively balanced, indicating even geographic penetration.

Insight 5: Sales distribution is right-skewed

Most orders are small-to-mid value. A handful of large Furniture orders significantly pull the mean above the median, indicating a few high-value transactions disproportionately contribute to total revenue.

Insight 6: Books and Sports are underperformers

Books (INR 0.25L) and Sports (INR 0.91L) contribute less than 7% of total revenue combined. Targeted promotions could help unlock growth in these segments.

Insight 7: Profit margin stands at 22.7%

Total profit of INR 4.15 Lakh against revenue of INR 18.29 Lakh gives an overall margin of 22.7%, which is healthy for an e-commerce model.

7. Conclusion

This project successfully demonstrates a complete end-to-end data analytics workflow using Python and MySQL. Starting from a raw dataset with duplicates and missing values, the data was cleaned, analysed statistically, visualised through an interactive multi-panel dashboard, and distilled into actionable business insights. Furniture and Electronics emerged as the key revenue drivers, while the East region showed the strongest geographic performance. The right-skewed distribution of sales values highlights the importance of premium product categories in overall revenue generation.

8. Future Improvements

- Implement time-series forecasting for monthly sales prediction.
- Add customer demographics for RFM (Recency, Frequency, Monetary) segmentation.
- Geographic heat-map visualisation using state-level data.
- Real-time dashboard integration with live data feeds.
- Predictive discount optimisation model.
- Automated weekly reporting pipeline.

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